

T3 BSM UV terms grouped by field content

EFT order: 5; loop order: 1. Columns are determined only from BSM UV terms. Empty cells denote no UV term with that field combination.

Field combinations 1 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	S_1	S_2	$F D$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{1}{2} N_{r_1}^T C N_{r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1} M_{S_1}^2$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$-M_{S_1}^2 \text{NewScalar1} \text{NewScalar1}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$+\frac{1}{2} i N_{r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{r_1})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{1}{2} N_{A_1, r_1}^T C N_{A_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1} M_{S_1}^2$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$-M_{S_1}^2 \text{NewScalar1} \text{NewScalar1}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$+\frac{1}{2} i N_{A_1, r_1}^T C \gamma^{\mu_1} D_{\mu_1} (N_{A_1, r_1})$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1} M_{S_1}^2$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$-M_{S_1}^2 \text{NewScalar1} \text{NewScalar1}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$+\overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1} M_{S_1}^2$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$ $-\text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$+\overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\overline{N_{a_1, r_1}} N_{a_1, r_1} M_F$	$-\text{NewScalar1} \text{NewScalar1} M_{S_1}^2$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$-M_{S_1}^2 \text{NewScalar1} \text{NewScalar1}$ $+\frac{1}{2} \text{Imbhd} \text{d} \text{Inv1} \text{NewScalar1} \text{NewScalar1} M_{S_1}^2$	$+\overline{N_{a_1, r_1}} \gamma^{\mu_1} D_{\mu_1} (N_{a_1, r_1})$

Field combinations 2 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	D	S_1	H	S_1	S_2	S_2	D
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}$	$(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{a_1})$	$+i\text{ambdaH1Inv1}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv1}_{a_1,a_2,a_2,a_4}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$ $+i\text{ambdaH1Inv2}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv2}_{a_1,a_2,a_2,a_4}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$+i\text{ambda11Inv1NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^\dagger\text{NewScalar1}_{a_2}^\dagger$ $+i\text{ambda11Inv2NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv2}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{a_1})$			
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}$	$(\text{NewScalar1}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{a_1})$	$+i\text{ambdaH1Inv1}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv1}_{a_1,a_2,a_2,a_4}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$ $+i\text{ambdaH1Inv2}H_{a_2}\text{NewScalar1}_{a_4}\text{T3HiggsPortalICGInv2}_{a_1,a_2,a_2,a_4}H_{a_1}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$+i\text{ambda11Inv1NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv1}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^\dagger\text{NewScalar1}_{a_2}^\dagger$ $+i\text{ambda11Inv2NewScalar1}_{a_4}\text{w5color1}_{a_4}\text{T1CrossScalarCGInv2}_{a_1,a_2,a_2,a_4}\text{w5color1}_{a_4}^\dagger\text{NewScalar1}_{a_2}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{a_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{a_1})$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1})^\dagger D_{\mu_1} (\text{NewScalar1})$	$+i\text{ambdaH1Inv1NewScalar1}H_{a_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2}\text{NewScalar1}^\dagger H_{a_1}^\dagger$	$+i\text{ambda11Inv1NewScalar1New8color1}_{a_2}\text{T1CrossScalarCGInv1}_{a_1,a_2}\text{NewScalar1}^\dagger\text{New8color1}_{a_1}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1})$			
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1})$	$+\frac{1}{2}i\text{ambdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$ $+\frac{1}{2}i\text{ambdaH1Inv2}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$ $+\frac{1}{2}i\text{ambdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2,A_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$ $+\frac{1}{2}i\text{ambdaH1Inv2}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv2}_{a_1,a_2,A_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$	$+i\text{ambda11Inv1NewScalar1New8color1}_{a_2}\text{T1CrossScalarCGInv1}_{a_1,a_2}\text{NewScalar1}^\dagger\text{New8color1}_{a_1}^\dagger$	$+D_{\mu_1} (\text{NewScalar2})^\dagger D_{\mu_1} (\text{NewScalar2})$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}$	$(\text{NewScalar1}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar1}_{A_1})$	$+\frac{1}{2}i\text{ambdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$ $+\frac{1}{2}i\text{ambdaH1Inv2}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$ $+\frac{1}{2}i\text{ambdaH1Inv1}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv1}_{a_1,a_2,A_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$ $+\frac{1}{2}i\text{ambdaH1Inv2}H_{a_2}\text{NewScalar1}_{A_2}\text{T3HiggsPortalICGInv2}_{a_1,a_2,A_2,A_1}H_{a_1}^\dagger\text{NewScalar1}_{A_1}^\dagger$	$+i\text{ambda11Inv1NewScalar1}_{a_2}\text{w5color1}_{a_2}\text{T1CrossScalarCGInv1}_{a_1,a_2,A_1,A_2}\text{w5color1}_{a_2}^\dagger\text{NewScalar1}_{a_1}^\dagger$ $+i\text{ambda11Inv2NewScalar1}_{a_2}\text{w5color1}_{a_2}\text{T1CrossScalarCGInv1}_{a_1,a_2,A_2,A_1}\text{w5color1}_{a_2}^\dagger\text{NewScalar1}_{a_1}^\dagger$ $+i\text{ambda11Inv1NewScalar1}_{a_2}\text{w5color1}_{a_2}\text{T1CrossScalarCGInv2}_{a_1,a_2,A_1,A_2}\text{w5color1}_{a_2}^\dagger\text{NewScalar1}_{a_1}^\dagger$ $+i\text{ambda11Inv2NewScalar1}_{a_2}\text{w5color1}_{a_2}\text{T1CrossScalarCGInv2}_{a_1,a_2,A_2,A_1}\text{w5color1}_{a_2}^\dagger\text{NewScalar1}_{a_1}^\dagger$	$+D_{\mu_1} (\text{NewScalar2}_{A_1})^\dagger D_{\mu_1} (\text{NewScalar2}_{A_1})$			

Field combinations 3 of 3

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2 H$	$S_1 F \ell$	$S_1 S_2 H$	$S_2 F \ell$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv1}_{a_1,a_2,a_3,a_4}H_{a_1}^{\dagger}\text{NewScalar2}_{a_2}^{\dagger}$ $+i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,a_3,a_4}H_{a_1}^{\dagger}\text{NewScalar2}_{a_2}^{\dagger}$	$+\overline{\ell_{a_1,i_1}}P_R N_{r_1}\text{NewScalar1}_{a_1}y_{1,i_1,r_1}$ $+\overline{y_{1,i_1,r_1}}N_{r_1}^T C P_L \ell_{a_1,i_1}\text{NewScalar1}_{a_1}^{\dagger}$	$+\overline{\lambda_{T3}}\text{NewScalar2}_{a_1}\epsilon_{a_2,a_3}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar1}_{a_3}^{\dagger}$ $+H_{a_1}H_{a_2}\text{NewScalar1}_{a_3}\lambda_{T3}\bar{\epsilon}_{a_2,a_3}\text{NewScalar2}_{a_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}N_{r_1}^T C P_L \ell_{a_1,i_1}\text{NewScalar2}_{a_2}\bar{\epsilon}_{a_1,a_2}$ $+\overline{\ell_{a_2,i_1}}P_R N_{r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{NewScalar2}_{a_1}^{\dagger}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv1}_{a_1,a_2,a_3,a_4}H_{a_1}^{\dagger}\text{NewScalar2}_{a_2}^{\dagger}$ $+i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar2}_{a_4}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,a_3,a_4}H_{a_1}^{\dagger}\text{NewScalar2}_{a_2}^{\dagger}$	$+\overline{\ell_{a_1,i_1}}P_R N_{A_1,r_1}\text{NewScalar1}_{a_2}y_{1,i_1,r_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2}$ $+\overline{y_{1,i_1,r_1}}N_{A_1,r_1}^T C P_L \ell_{a_2,i_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2}\text{NewScalar1}_{a_1}^{\dagger}$	$+\overline{\lambda_{T3}}\text{NewScalar2}_{a_1}\epsilon_{a_2,a_3}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar1}_{a_3}^{\dagger}$ $+H_{a_1}H_{a_2}\text{NewScalar1}_{a_3}\lambda_{T3}\bar{\epsilon}_{a_2,a_3}\text{NewScalar2}_{a_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}N_{A_1,r_1}^T C P_L \ell_{a_3,i_1}\text{NewScalar2}_{a_2}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}$ $+\overline{\ell_{a_3,i_1}}P_R N_{A_1,r_1}y_{2,i_1,r_1}\epsilon_{a_2,a_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_3,a_2}\text{NewScalar2}_{a_1}^{\dagger}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$ $+\frac{1}{2}i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$ $+\frac{1}{2}i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$ $+\frac{1}{2}i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$	$+\text{NewScalar1}\overline{\ell_{a_1,i_1}}C P_R \overline{N_{a_2,r_1}}^T y_{1,i_1,r_1}\epsilon_{a_1,a_2}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_1,i_1}^T C P_L N_{a_2,r_1}\bar{\epsilon}_{a_1,a_2}\text{NewScalar1}^{\dagger}$	$+\overline{\lambda_{T3}}\text{NewScalar2}_{A_1}\epsilon_{a_3,a_2}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}\text{NewScalar1}^{\dagger}H_{a_1}^{\dagger}H_{a_2}^{\dagger}$ $+\text{NewScalar1}H_{a_2}H_{a_3}\lambda_{T3}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar2}_{A_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}\overline{N_{a_1,r_1}}P_L \ell_{a_2,i_1}\text{NewScalar2}_{A_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2}$ $+\overline{\ell_{a_1,i_1}}P_R N_{a_2,r_1}y_{2,i_1,r_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2}\text{NewScalar2}_{A_1}^{\dagger}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+i\text{am bdaH2Inv1NewScalar2}H_{a_2}\text{T3HiggsPortal2CGInv1}_{a_1,a_2}\text{NewScalar2}^{\dagger}H_{a_1}^{\dagger}$	$+\overline{\ell_{a_3,i_1}}C P_R \overline{N_{a_1,r_1}}^T \text{NewScalar1}_{A_1}y_{1,i_1,r_1}\epsilon_{a_1,a_2}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_3,a_2}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_3,i_1}^T C P_L N_{a_1,r_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_2,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar1}_{A_1}^{\dagger}$	$+H_{a_2}H_{a_3}\text{NewScalar1}_{A_1}\lambda_{T3}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar2}^{\dagger}$ $+\text{NewScalar2}\overline{\lambda_{T3}}\bar{\epsilon}_{a_3,a_2}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar1}_{A_1}^{\dagger}$	$+\text{NewScalar2}y_{2,i_1,r_1}\overline{N_{a_1,r_1}}P_L \ell_{a_1,i_1}$ $+\overline{\ell_{a_1,i_1}}P_R N_{a_1,r_1}y_{2,i_1,r_1}\text{NewScalar2}^{\dagger}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$ $+\frac{1}{2}i\text{am bdaH2Inv1}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv1}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$ $+\frac{1}{2}i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$ $+\frac{1}{2}i\text{am bdaH2Inv2}H_{a_2}\text{NewScalar2}_{a_2}\text{T3HiggsPortal2CGInv2}_{a_1,a_2,A_1,A_2}H_{a_1}^{\dagger}\text{NewScalar2}_{A_1}^{\dagger}$	$+\overline{\ell_{a_3,i_1}}C P_R \overline{N_{a_1,r_1}}^T \text{NewScalar1}_{A_1}y_{1,i_1,r_1}\epsilon_{a_1,a_2}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_3,a_2}$ $+\overline{y_{1,i_1,r_1}}\ell_{a_3,i_1}^T C P_L N_{a_1,r_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_2,a_3}\bar{\epsilon}_{a_1,a_2}\text{NewScalar1}_{A_1}^{\dagger}$	$+\overline{\lambda_{T3}}\text{NewScalar2}_{a_2}\epsilon_{a_3,a_2}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}\text{gen}(\text{SU2L}(\text{fund}))_{A_2,a_1,a_2}H_{a_1}^{\dagger}H_{a_2}^{\dagger}\text{NewScalar1}_{A_1}^{\dagger}$ $-H_{a_2}H_{a_3}\text{NewScalar1}_{A_2}y_{1,i_1,r_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_3}\text{gen}(\text{SU2L}(\text{fund}))_{A_2,a_1,a_2}\bar{\epsilon}_{a_1,a_2}\text{NewScalar2}_{A_1}^{\dagger}$	$+\overline{y_{2,i_1,r_1}}\overline{N_{a_1,r_1}}P_L \ell_{a_2,i_1}\text{NewScalar2}_{A_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2}$ $+\overline{\ell_{a_1,i_1}}P_R N_{a_2,r_1}y_{2,i_1,r_1}\text{gen}(\text{SU2L}(\text{fund}))_{A_1,a_1,a_2}\text{NewScalar2}_{A_1}^{\dagger}$